

Array implementation of Stack

```
#include <stdio.h>

int stack[100],i,j,choice=0,n,top=-1;
void push();
void pop();
void show();
void main ()
{

    printf("Enter the number of elements in the stack ");
    scanf("%d",&n);
    printf("*****Stack operations using array*****");

    printf("\n-----\n");
    while(choice != 4)
    {
        printf("Chose one from the below options...\n");
        printf("\n1.Push\n2.Pop\n3.Show\n4.Exit");
        printf("\n Enter your choice \n");
        scanf("%d",&choice);
        switch(choice)
        {
            case 1:
            {
                push();
                break;
            }
            case 2:
            {
                pop();
                break;
            }
            case 3:
            {
                show();
                break;
            }
        }
    }
}
```

```

    }
    case 4:
    {
        printf("Exiting....");
        break;
    }
    default:
    {
        printf("Please Enter valid choice ");
    }
};
}
}

```

```

void push ()
{
    int val;
    if (top == n )
        printf("\n Overflow");
    else
    {
        printf("Enter the value?");
        scanf("%d",&val);
        top = top +1;
        stack[top] = val;
    }
}

```

```

void pop ()
{
    if(top == -1)
        printf("Underflow");
    else
        top = top -1;
}

void show()
{

```

```
for (i=top;i>=0;i--)
{
    printf("%d\n",stack[i]);
}
if(top == -1)
{
    printf("Stack is empty");
}
}
```